Jason Dsouza

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PROFESSIONAL SUMMARY

Geospatial developer with an M.Sc. in Geology and experience building Python data pipelines, spatial APIs and web-based R&D prototypes. Developed environmental intelligence and 3D hospital platforms at Genesys International, alongside GIS data engineering and cloud geospatial services at RadiXplore. Skilled in Python, FastAPI, Google Earth Engine, PostgreSQL/PostGIS and web-scraping, with hands-on experience in Three.js visualization and retrieval-augmented AI assistants.

TECHNICAL SKILLS

- **Programming and APIs:** Python, FastAPI, REST APIs, Basic JavaScript
- **GIS and Earth observation:** QGIS, GeoServer, Google Earth Engine Python API, remote sensing, spatial analysis
- **Geospatial processing:** GeoPandas, Shapely, Rasterio, GDAL, Fiona, OWSLib
- **Databases and deployment:** PostgreSQL, PostGIS, Docker, AWS EC2/S3, Linux, Git
- **3D and web visualization:** Three.js, MapLibre GL, CAD-to-GLB workflows, Streamlit, Folium
- **AI and automation:** RAG integration, Groq, Gemini API, OpenCV, Selenium, BeautifulSoup, Requests, Codex, Claude Code

PROFESSIONAL EXPERIENCE

Genesys International Corporation Ltd.

Apprentice - Trainee | *Research & Development, Middle East Division* | *Aug 2026 - Oct 2026*

- Developed EcoSentinel AI, a nature-reserve monitoring proof of concept, integrating environmental layers, wildlife and ranger-patrol monitoring, incident reporting and all-round surveillance.
- Built Python/FastAPI endpoints for vegetation indices, terrain and environmental data; integrated Google Earth Engine computation and weather APIs into an interactive Map Explorer.
- Implemented a retrieval-augmented generation (RAG) chatbot using Groq, alongside GPS simulation workflows for wildlife and patrol tracking and map-based incident registration.
- Built an AI Powered Smart Hospital proof of concept using CAD-derived data and GLB models; migrated visualization to Three.js.
- Developed prototype hospital modules for asset management, patient flow and AI assistance, incorporating feedback from stakeholder demonstrations.
- Containerized prototype builds with Docker in collaboration with the team; prepared technical notes and software requirements documentation to support reviews and delivery.

RadiXplore

Digital Earth Scientist (GIS) & Business Development Lead | *May 2023 - Jan 2025*

- Built Python pipelines to acquire, clean and standardize spatial datasets from government geological portals for database integration and downstream AI processing.
- Deployed and administered GeoServer WMS/WFS services on AWS EC2, backed by PostGIS for geometry storage and spatial queries.
- Validated spatial data with GeoPandas and Shapely, correcting geometry errors, projection inconsistencies and schema differences across datasets.
- Converted thousands of unstructured geological documents into structured geospatial layers by linking textual location references to spatial files.

- Automated data acquisition with Selenium, Requests and BeautifulSoup; documented Web-GIS setup and workflows in Confluence for non-GIS colleagues.
- Worked alongside the CEO for business development strategies and marketing campaigns.

PROJECTS

GeoBox-AR | *Python, OpenCV, Computer Vision, Hardware Integration, Linux*

- Designed and built a low-cost Augmented Reality Sandbox for the Geology Department at St. Xavier's College, currently in active use for student teaching.
- Developed the core software in Python using OpenCV (computer vision) to translate physical terrain interactions into real-time digital overlays, making geological concepts interactive without expensive commercial hardware.
- Used AI Agents like Claude Code for quick prototyping and frontend design
- Managed the full project scope including hardware sourcing, software development, faculty coordination, and deployment.
- Utilized open-source libraries including PySide6/PyQt (GUI), OpenCV, NumPy, and Libfreenect for data processing, making sure the software product remains cross-platform for Windows, Linux and MacOS users.
- Video Demo
- LinkedIn Article
- Blog Article
- Website (in progress)

Automatic Georeferencer | *Python, OpenCV, Multi-thread Computing*

- Built a completely offline automatic georeferencer for SOI 50K Toposheets, successfully georeferencing 250+ toposheets in under 7 minutes using parallel processing.
- Engineered an automated workflow using Pathlib and regex to index files, GeoPandas to load spatial boundaries, OpenCV to crop images to map frames, and Rasterio to assign coordinates to map corners. Planning to launch this as a QGIS plugin, soon.
- LinkedIn Article

A.I. Petrographer | *Python, Streamlit, Gemini-API*

- Built a web-app in Python where the user can upload Plane-Polarised and Cross-Polarised images of rock thin-sections to get A.I. insights about mineralogy, texture, rock name and more.
- Used Gemini API with a petrology-oriented system prompt for the LLM, Streamlit for GUI, file management and webhosting.
- Project Link

GIS Tools (Web-App) | *Python, Streamlit*

- Built a web-app in Python for on-the-go GIS visualisation and analysis.
- Current features include:-
    - Shapefile viewer that allows the user to drag and drop shapefiles, GeoJSONs and other spatial vector files onto a map and view them without having to load up desktop softwares.
    - Plot WMS layers
    - Download OSM features of an area by drawing a bounding box on a map
- Used Streamlit for webhosting, Folium for the map components, GeoPandas to handle vector data, OWSLib for fetching WMS layers.
- Project Link

The GIS Toolkit | *Blog Writing*

- Authored practical tutorials on open-source GIS workflows and spatial data engineering on a personal blog.
- Created content showing how geoscientists can transition from manual GIS tools to programmatic pipelines and leverage new advances in GIS and Geotech.
- Project Link

EDUCATION

Master of Science (M.Sc.) in Geology | St. Xavier's College, Mumbai | June 2024 - April 2026

- **Dissertation:** GeoBox-AR - Augmented reality framework for geoscience education.
- **Coursework:** GIS, remote sensing, Stratigraphy, Petrology, Economic Geology, Engineering Geology, Hydrogeology, Oceanography, Petroleum and Mining Geology

Bachelor of Science (B.Sc.) in Geology | St. Xavier's College, Mumbai | Aug 2020 - June 2023

CERTIFICATIONS

- ESRI Cartography Course (Esri)
- AI Aware & AI Appreciate Badges (AI For All)